

UA-UA Strategic Coordination Service Description Document

General Service Description							
Service description identification	Title: UA-UA Strategic Coordination Service Description Edition: v2.1.0 Reference Date: Sep 24, 2026						
Service identification	Name: ASTM F3548-21 Strategic Coordination Service						
Service abstract	This ASTM F3548-21 standard-based strategic coordination service is aimed at mitigating the worst credible outcome of collision between two or more UAS per FAA Order 8040.6A.						
Service Characteristics							
Service standard reference	Service standard reference: Standard Specification for UAS Traffic Management (UTM) UAS Service Supplier (USS) Interoperability (ASTM F3548-21) Implemented options of the service standard: - Conflicts are not permitted within the same priority level - Per FAA NTAP requirements, Service Providers shall retain data received from other service providers for a minimum period of 45 days.						
Operational Prioritization	<table border="1"> <thead> <tr> <th>Operation Type</th><th>ASTM F3548-21 Priority Level</th></tr> </thead> <tbody> <tr> <td>HIGH_PRIORITY</td><td>100</td></tr> <tr> <td>STANDARD</td><td>0</td></tr> </tbody> </table> A detailed description of operation type can be found in Operations Committee guidance on priority levels. Operators shall substantiate their eligibility for use of the high priority operation type per the criteria defined by the Operations Committee.	Operation Type	ASTM F3548-21 Priority Level	HIGH_PRIORITY	100	STANDARD	0
Operation Type	ASTM F3548-21 Priority Level						
HIGH_PRIORITY	100						
STANDARD	0						
High-level Description of Service Offer							
Operational environment	Refer to ASTM F3548-21 Sections 4.2.4-4.2.7 for operational needs addressed by the service and the capabilities offered by the service.						
Service functions	Refer to the following ASTM F3548-21 sections for functions offered						

	by the service and their real world effects. Sections 10.4, 10.5, 10.6, 10.7, 10.11, 10.12
Limitations and Constraints on Using the Service	
Service access and use conditions	Refer to the applicable UAS Service Provider Data Sharing and Governance Agreement and Operator - Service Provider Agreement
Security and Privacy Aspects	
Security	Refer to ASTM F3548-21 Sections 5.2 and A2 for security requirements. Relevant requirements are listed below: * GEN0005 - USSs performing any of the roles identified in this specification shall be implemented and operated under an ISO/IEC 27001-compliant Information Security Management System or equivalent. * DSS0200 - DSSs shall authenticate with other DSS instances in the same pool using an industry-standard authentication mechanism. * DSS0205 - Communication between DSSs shall be encrypted using an industry-standard encryption mechanism with a minimum encryption strength of 128 bits. Service Provider may demonstrate equivalent compliance to ISO/IEC 27001 by providing a written attestation that the Service Provider materially complies with ISO/IEC 27001 or has implemented an Information Security Management System that includes the following: • Risk Assessment. Annual information security risk assessment of UTM systems. • Access Control. The Service Provider will limit access to its UTM Service and Shared Data solely to Service Provider employees and contractors who need access to perform their duties. The Service Provider has sufficient controls in place to prevent unauthorized access to its UTM Services and Shared Data. At a minimum such controls will include multi-factor authentication, periodic review of access privileges, and immediate revocation of access privileges upon an employee's termination. • Data Control. Shared Data obtained in the operation of UTM Services is encrypted in transit. Shared Data is stored only on systems controlled by the Service Provider or via security-certified vendors. • Monitoring. Systems used in providing the UTM Services are monitored for configuration drift and unauthorized access. All employee endpoints run malware screening and remediation tooling. • Incident Response. Established process for incident response and breach notification.
Privacy	Refer to ASTM F3548-21 Sections 5.2 and A2 for security requirements. Relevant requirements are listed below:

	* USSs performing any of the roles identified in this specification shall (GEN0010) be implemented and operated under an ISO/IEC 27701-compliant Privacy Information Management System or equivalent. For initial production flights, Service Providers may demonstrate equivalent compliance for ISO/IEC 27701 as required by ASTM F3548-21 as follows: • A Service Provider conforms with the terms set forth in Annex D (Data Privacy) • A Service Provider completes an annual self-assessment and makes a written attestation during onboarding that the Service Provider materially complies or has implemented the following: ○ Conduct annual privacy risk assessment. ○ Implement privacy controls as part of a Privacy Information Management System (PIMS). ○ Implement privacy and security roles and responsibilities. ○ Define and adhere to a consent management process. ○ Manage Data Subject Rights (including as set forth in Annex D). ○ Establish processes for incident response and breach notification.
Authentication	Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0. Auth provider should be ISO27001 certified or approved equivalent
Quality Aspects	
Quality of service	Refer to ASTM F3548-21 Section 5.2 for quality of service requirements. Relevant requirements are listed below: *GEN0015 - USSs performing any of the roles identified in this specification shall be implemented and operated under an ISO/IEC 9001-compliant Quality Management System, or equivalent. Service providers may demonstrate equivalent compliance to ISO/IEC 9001 by providing a written attestation that the Service Provider materially complies with ISO/IEC 9001 or has implemented a quality management system that includes the following: • Quality policy • Procedures to identify and document key processes and procedures for maintaining, testing, and updating the UTM Services

	• Process for identifying, investigating, and addressing quality issues and implementing corrective and preventive actions • Criteria for selecting and monitoring suppliers and subcontractors to ensure they meet quality requirements • Plan for fostering continual improvement that includes employee feedback and initiatives to enhance effectiveness and efficiency of processes.
Service verification information	The Strategic Coordination Service including associated DSS interoperability will be verified prior to SW deployment using test baseline identifier TB-8fd8148
Discovery and Synchronization Service (DSS) pool configuration	DSS configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance. Requirements for confirming DSS pooling integrity Upon joining a pool, a USS hosting a DSS instance shall verify the Dynamic Airspace Representation Identifier matches the published Dynamic Airspace Representation Identifier for the pool. The first instance of the DSS will provide the published Dynamic Airspace Representation Identifier for other pooling entities. USS shall continuously monitor that Dynamic Airspace Representation Identifier for their DSS instances match the DSS pool. Continuously is defined as at least once every 24 hours. For CockroachDB implementation, USS shall notify when pool health metrics are not in bounds within 1 minute of occurrence. In bound health metrics include: • Unavailable Range = 0 Under nominal operating conditions, unavailability is indicative of cluster misconfiguration. • SQL Failure Count = 0 SQL Failure count is a rare but serious indication of a malformed query. Alerting on it in earlier environments helps ensure that an incorrect DSS version update never reaches production. • Active node metric absence for a rolling 5 minutes window. Metric absence indicates that the state of the cluster is unknown and indicates an inability to monitor critical metrics of the system. • Node restarts ≤ 1 within an hour. Frequent restarts indicate instability. Requirements for DSS eviction procedures • A responsible USS, designated by the Technical Committee,

	shall periodically cleanup expired Strategic Conflict Detection entities including operational intent references and subscriptions from the DSS pool's shared CockroachDB cluster per the Technical Committee DSS eviction procedures. Requirements for Datastore Connection Pool Sizing Each USS hosting a DSS instance shall configure the `datastore_max_open_conns` parameter to the value specified by the Technical Committee.								
Safety	Service providers will conduct a hazard analysis to identify and mitigate safety risks in accordance with the process outlined by the GUTMA Safety Task Force Paper "USSP's Safety Support Assessment"								
Performance Requirements									
Nominal service behavior	Refer to the attached Requirements Traceability Matrix, <i>Strategic Coordination Compliance Matrix V2.0</i>, for service performance and verification requirements								
Degraded service behavior	Notifications regarding degraded service behavior including DSS degradation will be sent per the terms in the applicable UAS Service Provider Data Sharing and Governance Agreement and Operator-Service Provider agreement. Mitigation of a Severity 1 issue as defined in the applicable UAS Service Provider Data Sharing and Governance Agreement includes a service provider marking their own availability state as down in the DSS until the issue is resolved. DSS-related service degradation detected per the requirements for confirming pooling integrity and provision of DSS instances are Severity 1 events.								
Operational Intent	Operational intents ensure 95% conformance and are developed per the standard								
DSS version and configuration	To participate in the deployment of a DSS pool, a USP must meet the following version and configuration requirements: <table border="1"> <thead> <tr> <th>Setting / Configuration</th><th>Required value</th></tr> </thead> <tbody> <tr> <td>DSS version</td><td>InterUSS DSS v0.23</td></tr> <tr> <td>SCD global lock</td><td>OFF</td></tr> <tr> <td>SCD hash lock</td><td>OFF</td></tr> </tbody> </table>	Setting / Configuration	Required value	DSS version	InterUSS DSS v0.23	SCD global lock	OFF	SCD hash lock	OFF
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DSS version	InterUSS DSS v0.23								
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	Time-based notification index	ON
	CRDB SCD schema version	3.4.1
	CRDB RID schema version	4.1.0
DSS infrastructure requirements	To participate in the deployment of a DSS pool, a USP must meet the following minimum infrastructure requirements: Configuration ○ Provide exactly 3 CockroachDB nodes ○ Each node must have at least ■ DNS resolution or stable public static IP to enable node discovery ■ Port 26257 availability for inter-node communication ■ CockroachDB version v24.1.3 Provision of DSS instances These requirements ensure effective minimal provision of DSS instances. It is also recommended that Service Providers adhere to InterUSS recommended sizing and proactively monitor conditions at lower levels to reduce the likelihood of exceeding these requirements. ○ USS shall provision DSS instances with the necessary resources to maintain CPU usage of < 90%. ○ USS shall notify if CPU usage >=90% for more than 1 consecutive hour. ○ USS shall provision DSS instances with the necessary resources to maintain memory usage of < 90%. ○ USS shall notify if memory usage >=90% for more than 1 consecutive hour. ○ USS shall provision DSS instances with the necessary resources to maintain IOPS in progress < 10. ○ USS shall notify if IOPS in progress >=10 for more than 5 consecutive minutes. ○ USS shall provision DSS instances with the necessary resources to maintain <=100ms P50 Round Trip Time (RTT) between DSS instances. 100ms is based on the nominal performance for inter cloud RTT times within the continental US as an initial target value. ○ USS shall notify if P50 Round Trip Time (RTT) > 100ms more than 5 consecutive minutes. ○ For CRDB implementations, the following definitions apply	
	Requirement	CRDB metric per CRDB node

	CPU Usage	sys.cpu.combined.percent-normalized
	Memory	sys.rss
	IOPS in progress	sys.host.disk.iopsinprogress
	Storage	capacity.used
	P50 Round Trip Time (RTT)	round-trip-latency (50 Percentile)
	As of the reference date of this SDD, the following configuration generally meets the above Provision of DSS instances requirements ○ CPU: At least 4 cores; 8+ cores recommended, with support for virtualization and containerization. ○ RAM: At least 8GB of memory; 16GB+ recommended. ○ Storage: SSD-based storage with at least 100GB available per node (expandable based on dataset size and replication needs). ■ Minimum IO/s 3000 R/W ■ Minimum throughput 125 MB/s R/W ■ Maximum average latency 5ms R/W ■ Maximum P99 R/W latency 10ms ○ Network: High-speed (10+Gbps) low-latency network with a stable public IP address ○ Accurate system time using NTP synchronization from Google's Public NTP source	